SIH 2026 EXECUTIVE CHEAT SHEET

Exactly What We Are Predicting

Quick reference guide on the ML Target, Input Features, and the 4 Landed Cost Heads for SAIL

Ministry of Steel / SAIL
Overseas Bulk Coal Logistics

ML TARGET The Future Ocean Freight Rate (Y)

Zero Data Leakage • 15-Day Forward Lookahead

The Machine Learning model predicts **EXACTLY ONE THING**: the market spot price to charter a bulk carrier vessel **15 trading days in the future**, expressed in **US Dollars per Metric Ton ($/MT)** for carrying coking coal from **Gladstone (Australia) to East Coast India (Haldia/Paradip)**.

VARIABLE SYMBOL	UNIT OF MEASUREMENT	FORECASTING HORIZON	VALIDATED TEST ERROR
Y (Target)	$/Metric Ton	15 Trading Days	±$0.716 / MT
target_freight_rate_proxy(t + 15d)	Typically $9.00 to $14.50 / MT	Lookahead window for chartering	6.68% MAPE on untouched holdout

THE 3 QUANTILE SCENARIOS (P10, P50, P90)

P10: Optimistic Case — 10th Percentile

Low Market Rate: There is only a 10% chance the market rate will drop below this. Used for best-case budget lower bound.

P50: Expected Base Case — Median

Expected Most Likely Rate: The primary forecast used to compare against today's spot rate for decision making.

P90: Worst Case / Spike — 90th Percentile

Pessimistic Market Spike: 90% chance the rate stays below this. Critical for SAIL risk hedging against sudden freight surges.

What ML Predicts vs. What Are Inputs (X)

Item / Variable	Role in System	Why?
Ocean Freight ($/MT)	ML PREDICTED (Y)	Volatile ocean charter market.
Baltic Index (BDRY)	Input Feature (X)	Historical leading indicator.
Coal Price (FRED)	Input Feature (X)	Exogenous commodity demand.
Bunker Fuel ($/MT)	Input Feature (X)	Ship operating cost driver.
USD / INR Exchange	Input Feature (X)	Currency macro exposure.

What Is Calculated by Math Formula

Cost Head	Calculation Method	Formula Basis
Voyage Days	Physics Math	Distance (NM) ÷ (Speed × 24)
Fuel Burn Cost	Physics Math	Days × Daily Burn MT × $/MT
Port Demurrage	Contract Math	(Wait Days - Laytime) × Penalty $/Day
Sandheads Lightering	Port Tariff Math	Offloaded MT × $7.00/MT (Haldia)
Total Landed Cost	Deterministic Sum	Sum of all 4 cost heads ÷ Cargo MT

THE 4 LANDED COST HEADS PAID BY SAIL

COST HEAD #1

Base Ocean Freight

`Cargo MT × Predicted $/MT`

The core charter fee paid to shipowner. **Uses the ML predicted freight rate!**

COST HEAD #2

Voyage Bunker Fuel

`Days × MT/Day × Bunker $/MT`

Marine fuel (VLSFO) burned during transit (e.g., 5,250 NM at 12.5 knots = 17.5 days).

COST HEAD #3

Port Demurrage Fine

`Excess Wait Days × $/Day`

Congestion penalty paid if vessel waits at port anchorage longer than contract laytime.

COST HEAD #4

Sandheads Lightering

`Lightered MT × $7.00/MT`

Mandatory at Haldia (8.5m draft) for large Panamax vessels (14.5m draft) before entering port.

THE 2 CORE DECISIONS DELIVERED TO SAIL PROCUREMENT DESK

Decision 1: Optimal Vessel Size

Evaluates Handysize (35k MT), Supramax (55k MT), and Panamax (75k MT) by calculating the **Total Landed Cost per MT** across all 4 cost heads. It recommends the cheapest vessel class after factoring in Haldia's shallow draft lightering fees vs. bulk discounts.

Decision 2: Charter Timing (FIX NOW vs HOLD)

Compares **Today's Spot Freight Rate** against the **15-day Forward Forecast (P50)**:

`FIX NOW: Today's Rate < Forecast P50 (Lock in today before rates rise).`

`HOLD: Today's Rate > Forecast P50 (Wait 15 days because rates will drop).`